

Math 445/545. Winter 2026. Practice problems for the Final.

The problems below illustrate various techniques you are supposed to know for the final. Some of the problems below are more difficult than the ones you will see on the final.

Problem 0. Please review Midterm 1, Midterm 2 (including practice problems).

Problem 1. Find the group $\mathbb{Z}^3/H$ where H is generated by $(6, 8, 12)$ and $(4, 10, 14)$.

Solution. Using elementary matrix operations we get

$$\begin{pmatrix} 6 & 8 & 12 \\ 4 & 10 & 14 \end{pmatrix} \sim \begin{pmatrix} -2 & 8 & 12 \\ -6 & 10 & 14 \end{pmatrix} \sim \begin{pmatrix} -2 & 0 & 0 \\ -6 & -14 & -22 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 \\ 0 & 8 & -22 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}.$$

Answer: $\mathbb{Z}^3/H \simeq \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}$.

Problem 2. Let A be a square integral $n \times n$ matrix and let $B = \mathbb{Z}^n/H$ where H is generated by the rows of A . Prove that B is finite if and only if $\det A \neq 0$. If B is finite prove that order of B is absolute value of $\det A$.

Solution. We know that using elementary matrix operations matrix A can be transformed into diagonal matrix $\text{diag}(d_1, \dots, d_n)$. The group $B = \mathbb{Z}^n/H$ is infinite if and only if $d_i = 0$ for some i ; if all d_i are nonzero then $|B| = |d_1 \cdots d_n|$. Now the result follows since the elementary matrix operations do not change $|\det A|$.

Problem 3. Prove that the group $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$ is cyclic if and only if a and b are relatively prime. How to find a generator of this group?

Solution. First of all notice that the group $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$ is infinite. This is clear if $a = b = 0$; if say $a \neq 0$, then we have infinitely many distinct cosets of the form $(0, n) + \langle(a, b)\rangle$ (for $n \in \mathbb{Z}$). Thus $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$ is cyclic if and only if it is isomorphic to $\mathbb{Z}$.

Assume that $\gcd(a, b) = r > 1$. Then $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$ contains an element of order r , namely $(\frac{a}{r}, \frac{b}{r}) + \langle(a, b)\rangle$, so it can't be isomorphic to $\mathbb{Z}$.

Assume that $\gcd(a, b) = 1$. Then we can find integers p and q such that $pa - qb = 1$. We claim that the coset $x = (q, p) + \langle(a, b)\rangle$ is a generator of $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$. Indeed, the subgroup of $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$ generated by x contains both cosets $(1, 0) + \langle(a, b)\rangle$

(since $-b(q, p) = (1, 0) - p(a, b)$) and $(0, 1) + \langle(a, b)\rangle$ (since $a(q, p) = (0, 1) + q(a, b)$), so it must coincide with $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$.

Problem 4. Let N be a subgroup of $\mathbb{Z}^{10}$ generated by 7 elements. Prove that $\mathbb{Z}^{10}/N$ is infinite.

Solution. Let us write 7 generators as a 10×7 matrix. Its Smith normal form will have 3 or more zero columns. Thus $\mathbb{Z}^{10}/N$ will have at least 3 infinite cyclic groups in its decomposition; certainly this implies that $\mathbb{Z}^{10}/N$ is infinite.

Problem 5. (a) What is the number of ideals in the ring $\mathbb{R}[x]/(x^3 + x + 1)$?
(b) What is the number of ideals in the ring $\mathbb{Z}/1000\mathbb{Z}$?

Solution. (a) The ideals in $\mathbb{R}[x]/\langle x^3 + x + 1 \rangle$ are in bijection with ideals in $\mathbb{R}[x]$ containing $\langle x^3 + x + 1 \rangle$. Such ideals are precisely principal ideals $\langle p(x) \rangle$ where $p(x)$ is a divisor of $x^3 + x + 1$. The polynomial has just one real root (since its derivative is always positive), so it factorizes into 2 irreducible factors. Thus the number of divisors is $2^2 = 4$.

Answer: 4 ideals (including the trivial and improper ones).

(b) Similarly to previous problem we just need to count the number of divisors of 1000. Since $1000 = 2^3 \cdot 5^3$ we have $4 \cdot 4 = 16$ divisors.

Answer: 16 ideals.

Problem 6. What is the number of irreducible polynomials of the form $x^2 + ax + b$ in the ring $\mathbb{Z}_p[x]$? (Hint: First find the number of reducible polynomials of this form.)

Solution. We have p options for a and p options for b , so we have p^2 polynomials of the form $x^2 + ax + b$. The reducible polynomials are of the form $(x - \alpha)(x - \beta)$ with $\alpha, \beta \in \mathbb{Z}_p$; there are $\frac{p(p-1)}{2}$ of such polynomials with $\alpha \neq \beta$ (this is the number of 2-element subsets of $\mathbb{Z}_p$) and p more polynomials with $\alpha = \beta$. The remaining $p^2 - \frac{p(p-1)}{2} - p = \frac{p(p-1)}{2}$ polynomials are irreducible.

Answer: there are $\frac{p(p-1)}{2}$ irreducible polynomials of the form $x^2 + ax + b$ in $\mathbb{Z}_p[x]$.

Problem 7. Are the rings $\mathbb{Z}[x]/\langle x^2 - 1 \rangle$ and $\mathbb{Z} \oplus \mathbb{Z}$ isomorphic?

Solution. Assume $\phi : \mathbb{Z}[x]/\langle x^2 - 1 \rangle \rightarrow \mathbb{Z} \oplus \mathbb{Z}$ is an isomorphism. Then $\phi(1)$ is the identity, so $\phi(1) = (1, 1)$. Also ϕ send the coset of x to an element y of $\mathbb{Z} \oplus \mathbb{Z}$ such that y^2 is the identity. Hence we have 4 options for y : it must be one of the elements $(\pm 1, \pm 1)$. Note that the (additive) abelian group $\mathbb{Z}[x]/\langle x^2 - 1 \rangle$ is generated by the cosets of 1 and x (e.g. coset of $2x^3 - x^2$ is the same as coset of $2x - 1$ since $2x^3 - x^2 = 2x(x^2 - 1) + 2x - (x^2 - 1) - 1 = 2x - 1 \pmod{\langle x^2 - 1 \rangle}$). Thus if ϕ is surjective then $\mathbb{Z} \oplus \mathbb{Z}$ is generated as an abelian group by $\phi(1)$ and y . However this is not the case: any integer linear combination of $(1, 1)$ and $(\pm 1, \pm 1)$ will have both coordinates of the same parity, so $(1, 0)$ is not in the image of ϕ .

Answer: the rings $\mathbb{Z}[x]/\langle x^2 - 1 \rangle$ and $\mathbb{Z} \oplus \mathbb{Z}$ are not isomorphic.

Problem 8. Let W be a 6-dimensional vector space.

a) Prove that for any 4-dimensional subspaces $W_1, W_2 \subset W$ the intersection $W_1 \cap W_2$ contains a non-zero vector.

b) Is it true that the intersection $W_1 \cap W_2 \cap W_3$ of any three 4-dimensional subspaces $W_1, W_2, W_3 \subset W$ contains a nonzero vector?

Solution. a) Let v_1, v_2, v_3, v_4 be a basis of W_1 and let w_1, w_2, w_3, w_4 be a basis of W_2 . Then 8 vectors $v_1, v_2, v_3, v_4, w_1, w_2, w_3, w_4$ are linearly dependent since the dimension of W is $6 < 8$. Thus we have linear combination

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \alpha_4 v_4 + \beta_1 w_1 + \beta_2 w_2 + \beta_3 w_3 + \beta_4 w_4 = 0$$

where not all coefficients are zero. Equivalently,

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \alpha_4 v_4 = x = -(\beta_1 w_1 + \beta_2 w_2 + \beta_3 w_3 + \beta_4 w_4).$$

Clearly $x \in W_1 \cap W_2$ and $x \neq 0$ since v_i and w_i are linearly independent.

b) Not true. For example, take $W = k^6$, $W_1 = \{(x_1, x_2, x_3, x_4, 0, 0), x_i \in k\}$, $W_2 = \{(0, 0, y_1, y_2, y_3, y_4), y_i \in k\}$, and $W_3 = \{(z_1, z_2, 0, 0, z_3, z_4), z_i \in k\}$. Then $W_1 \cap W_2 \cap W_3 = 0$.

Problem 9. Let V be a n -dimensional vector space over the finite field $\mathbb{Z}_p$.

a) How many vectors are in V ?

b) How many 1-dimensional subspaces does V have?

Solution: a) Let $e_1, \dots, e_n$ be a basis for V . Then any vector $v \in V$ can be uniquely represented in the form $v = \sum_i a_i e_i$, where $a_1, \dots, a_n \in \mathbb{Z}_p$. Thus V has p^n elements.

b) Let P denote the set of 1-dimensional subspaces of V . Define a map $\psi : V - 0 \rightarrow P$ to be $\psi(v) = \text{span } v$. Note that ψ is surjective: any 1-dimensional subspace is generated by a single vector. Observe, that $\psi(v) = \psi(w)$ if and only if $v = cw$, for some nonzero scalar $c \in \mathbb{Z}_p^*$. Thus the preimage of any element $l \in P$ consists of $p - 1$ vectors. Therefore $|V - 0| = (p - 1) \times |P|$ and $|P| = \frac{p^n - 1}{p - 1} = 1 + p + \dots + p^{n-1}$.

Problem 10. How many elements are in $\mathbb{Z}[\sqrt{-2}]/\langle 7 + 5\sqrt{-2} \rangle$?

Solution. The ideal $\langle 7 + 5\sqrt{-2} \rangle$ consists of all multiples $(7 + 5\sqrt{-2})(x + y\sqrt{-2}) = x(7 + 5\sqrt{-2}) + y(-10 + 7\sqrt{-2})$ with $x, y \in \mathbb{Z}$. In other words the ideal $\langle 7 + 5\sqrt{-2} \rangle$ is abelian subgroup of $\mathbb{Z}[\sqrt{-2}] \simeq \mathbb{Z}^2$ generated by $7 + 5\sqrt{-2}$ and $-10 + 7\sqrt{-2}$. In matrix form we get $\begin{pmatrix} 7 & 5 \\ -10 & 7 \end{pmatrix}$. Now we get from Problem 2 that $\mathbb{Z}[\sqrt{-2}]/\langle 7 + 5\sqrt{-2} \rangle$ is finite and has precisely $7 \cdot 7 - 5 \cdot (-10) = 99$ elements.

Answer: the ring $\mathbb{Z}[\sqrt{-2}]/\langle 7 + 5\sqrt{-2} \rangle$ has 99 elements.

Problem 11. Let $0 \neq a + bi \in \mathbb{Z}[i]$. Prove that the ring $\mathbb{Z}[i]/\langle a + bi \rangle$ is finite and find its cardinality.

Solution: The ideal $\langle a + bi \rangle$ consists of all multiples $(a + bi)(x + yi) = x(a + bi) + y(-b + ai)$ with $x, y \in \mathbb{Z}$. In other words the ideal $\langle a + bi \rangle$ is abelian subgroup of $\mathbb{Z}[i] \simeq \mathbb{Z}^2$ generated by $a + bi$ and $-b + ai$. In matrix form we get $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Now we get from Problem 3 that $\mathbb{Z}[i]/\langle a + bi \rangle$ is finite if and only if $a^2 + b^2 \neq 0$ and has precisely $a^2 + b^2$ elements in this case.

Answer: the ring $\mathbb{Z}[i]/\langle a + bi \rangle$ has $a^2 + b^2$ elements (provided that $a + bi \neq 0$).

Problem 12. (a) Prove that $\mathbb{Z}[x]/(x^5 + 3x - 5)$ is finitely generated when considered as an abelian group.

(b) Prove that $\mathbb{Z}[x]/(2x^2 + 1)$ is not finitely generated when considered as an abelian group.

Solution: (a) Using long division we see that each element of $\mathbb{Z}[x]/(x^5 + 3x - 5)$ can be represented by polynomial with integer coefficients of degree ≤ 4 and this representation is unique. In other words the abelian group $\mathbb{Z}[x]/(x^5 + 3x - 5)$ is isomorphic to the abelian group of polynomials with integer coefficients of degree ≤ 4 which is in turn isomorphic to $\mathbb{Z}^5$ and is finitely generated.

(b) Assume the contrary. Let N be the largest degree of the generators of $\mathbb{Z}[x]/(2x^2 + 1)$. Then x^{N+1} can't be represented as a linear combination of the generators plus element of the ideal $(2x^2 + 1)$ since the leading coefficient of any element of $(2x^2 + 1)$ is divisible by 2. In other words the coset $x^{N+1} + (2x^2 + 1)$ can't be expressed as a linear combination of generators. This is a contradiction.

Remark. Generalization: The ring $\mathbb{Z}[x]/(p(x))$ is finitely generated as an abelian group if and only if the leading coefficient of $p(x)$ is ± 1 . Prove this!